UNIVERSITY EXAMINATION BOARD

ADVANCED CALCULUS & DIFFERENTIAL EQUATIONS

Examination Year: 2026 | Maximum Marks: **100** | Total Questions: **15** | Time Allowed: **3 Hours**

GENERAL INSTRUCTIONS:

1. All questions are compulsory unless specified otherwise.
2. Candidates must show all working, intermediate steps, and derivations to earn full marks.
3. Draw diagrams wherever necessary; label all parts clearly.
4. Figures, graphs and diagrams are provided with the respective question.
5. Use of scientific calculator is permitted unless otherwise notified.

Q.No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total
Marks	5	7	6	7	7	6	7	7	7	6	7	7	6	7	7	100

Question 1 **[5 Marks]**

Compute the directional derivative of f(x,y,z) = x^2 y + y^2 z - z^2 x at point (1,1,1) in the direction of **v** = (2,-1,2).

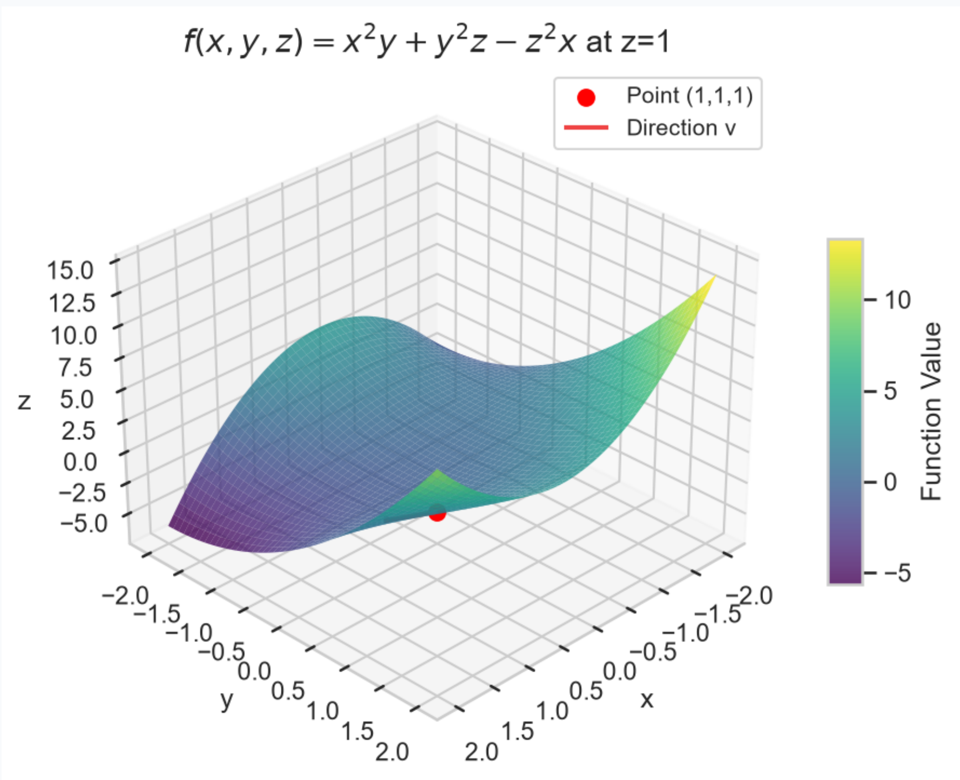

Figure 1

Question 2 [7 Marks]

Evaluate the surface integral _S (x^2 + y^2) dS where S is the hemisphere x^2 + y^2 + z^2 = 4 with z ≥ 0.

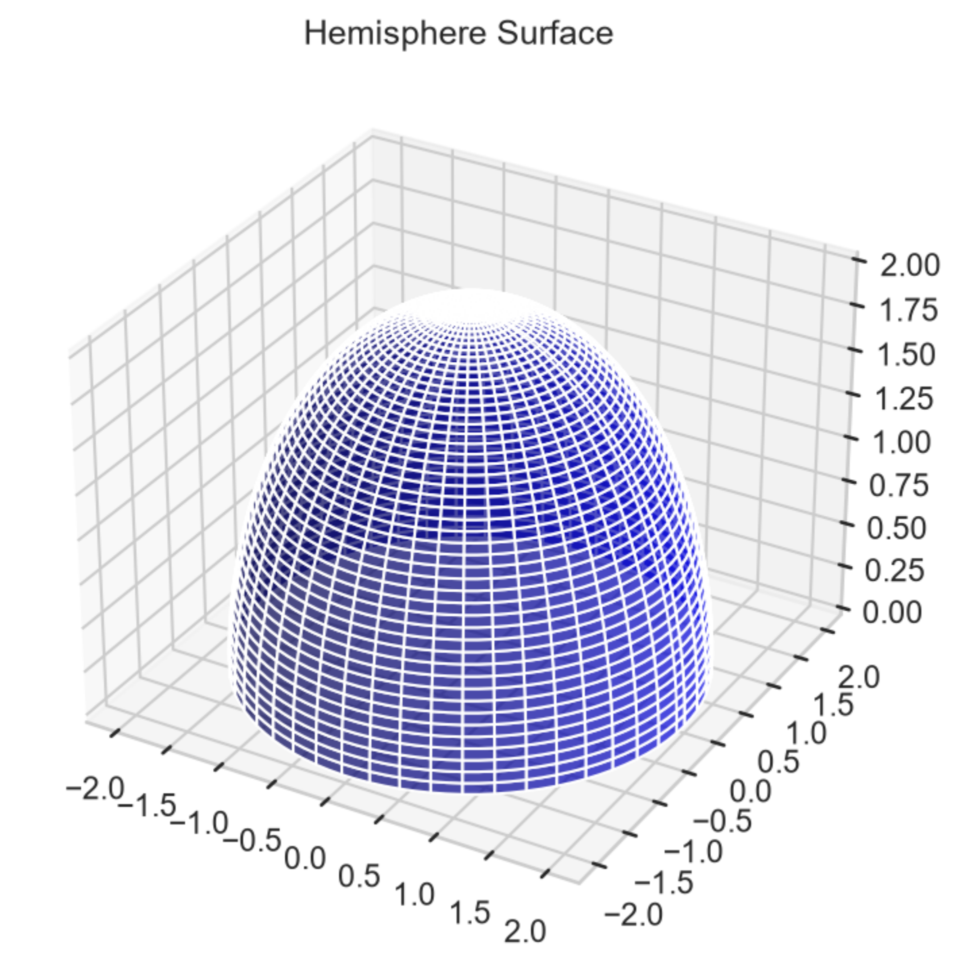

Figure 2

Question 3 [6 Marks]

Solve the differential equation (dy / dx) + y tan x = sin 2x with initial condition y(0) = 1.

Question 4 [7 Marks]

Find the critical points of f(x,y) = x^3 + y^3 - 3xy and classify them as maxima, minima, or saddle points.

Question 5

[7 Marks]

Calculate the flux of **F** = (x^2, y^2, z^2) through the cube $[0,1] \times [0,1] \times [0,1]$.

Question 6

[6 Marks]

Solve the exact differential equation $(2xy + y^3)dx + (x^2 + 3xy^2)dy = 0$.

Question 7

[7 Marks]

Compute the line integral $\int_C (x^2 + y^2)\, ds$ where C is the circle $x^2 + y^2 = 9$ traversed counterclockwise.

Question 8

[7 Marks]

Find the general solution to $y'' - 4y' + 4y = e^{2x} \cos x$.

Question 9

[7 Marks]

Evaluate _V (x^2 + y^2) dV where V is the region bounded by z = x^2 + y^2 and z = 4.

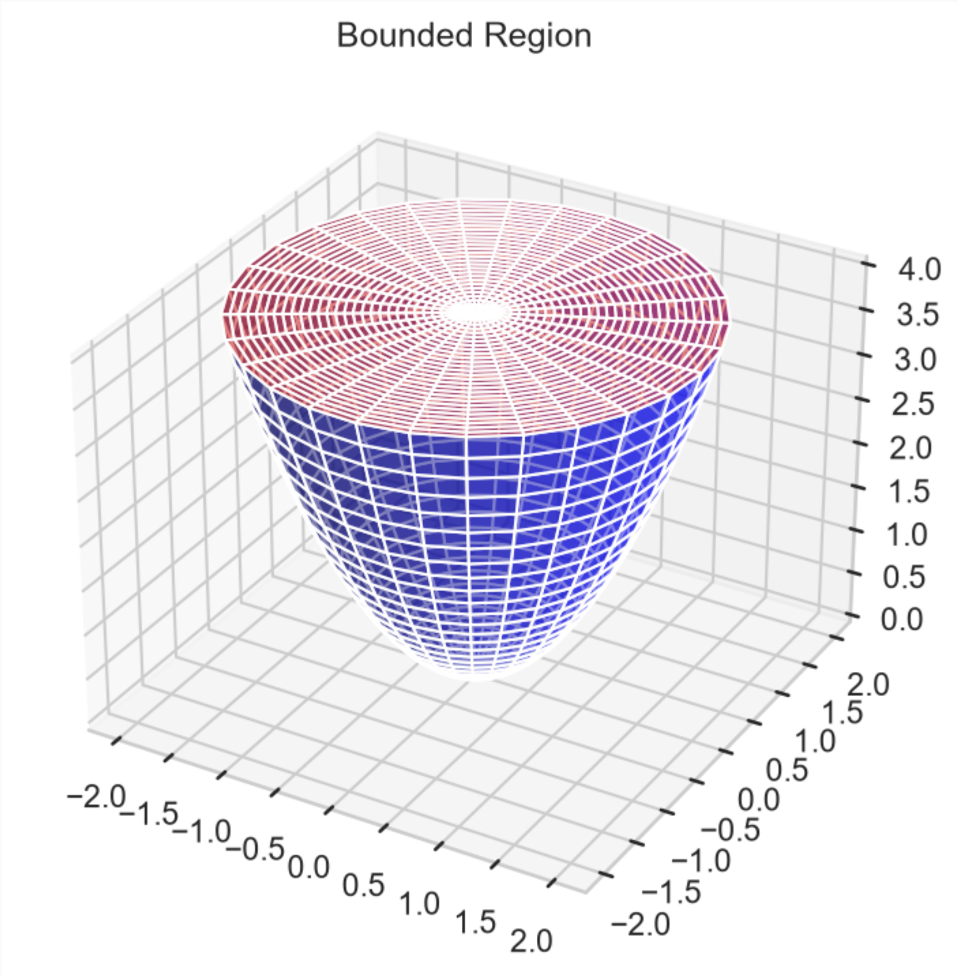

Figure 3

Question 10

[6 Marks]

Solve the Bernoulli equation (dy / dx) + (y / x) = x^2 y^3.

Question 11

[7 Marks]

Find the work done by **F** = (y^2, x^2, z) along the helix **r**(t) = (cos t, sin t, t) from t=0 to t=2π.

Question 12

[7 Marks]

Determine if **F** = $(2xy + z^3, x^2, 3xz^2)$ is conservative and find its potential function if it exists.

Question 13

[6 Marks]

Solve the initial value problem $y' = (x^2 + y^2 / xy)$ with $y(1) = 2$.

Question 14

[7 Marks]

Compute the surface area of $z = \sqrt{(x^2 + y^2)}$ for $1 \leq x^2 + y^2 \leq 4$.

Question 15

[7 Marks]

Find the general solution to the system: $(dx / dt) = 2x + y$, $(dy / dt) = -x + 4y$.